TECHNICAL TEST SUITE EXECUTION

AI Weather Wise — API Testing Report

Comprehensive validation of authentication, location-based weather tracking, and Gemini AI weather intelligence services.

Testing Objective & Environment

Objective: Verify authentication, protected routes, location/weather query management, and Gemini AI integration.

Stack: Node.js + Express.js, MongoDB

Testing Tools: Thunder Client (VS Code) / Postman

Security: Bearer JWT Authentication

AI Provider: Google Gemini API

Base Server URL: `http://localhost:5000`

1. Authentication & Authorization Tests

- `POST` `/api/auth/register` — Valid user creation with email, password, and home location preference.
- `POST` `/api/auth/login` — Valid credentials return a signed JWT token and user profile details.
- `GET` `/api/auth/profile` — Retrieve authorized user profile using valid Bearer JWT.
- **Negative Tests:** Reject duplicate emails, invalid credentials, and requests with missing or expired Bearer tokens.

2. Weather Search & Saved Locations Tests

- `POST` `/api/weather/locations` — Save a favorite location (city, lat/lon, alert preferences).
- `GET` `/api/weather/locations` — Fetch all saved locations for the authenticated user.
- `GET` `/api/weather/locations/:id` — Retrieve specific location details and stored forecast history.
- `PUT` `/api/weather/locations/:id` — Update alert thresholds (e.g., rain notifications, temperature units).
- `DELETE` `/api/weather/locations/:id` — Remove a saved location from the user profile.
- `GET` `/api/weather/search` — Query live weather data by city name, postal code, or geographic coordinates.
- **Validation Checks:** Enforce required fields, coordinate ranges (lat -90 to 90, lon -180 to 180), and valid temperature units (`Celsius` / `Fahrenheit`).

3. Gemini AI Intelligence Endpoint Tests

- `POST` `/api/ai/forecast-advisory` — Inputs: *location, activityType, forecastData*; Expected: Personalised outfit suggestions, outdoor activity recommendations, and weather safety warnings.
- `POST` `/api/ai/climate-insights` — Inputs: *historicalTemp, precipitationTrends, location*; Expected: Seasonal performance analysis, anomalies summary, actionable preparation tips, and tailored alerts.

Expected Results & Test Evidence Summary

All valid operations must return 2xx HTTP success codes accompanied by structured JSON payloads. Authentication, input validation, database query, and third-party AI integration errors must generate structured JSON error models with clear diagnostic messages.

Required Evidence: Screenshots capturing response payloads for Registration, Login, Profile Retrieval, Weather